

# Deceptive density: evidence-oriented inference reveals how subsidies can mask predator death-traps

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## Introduction

This analysis examines Mabuya (*Trachylepis atlantica*; skink) populations across three distinct sites (i.e., LPU [Low Predator Unsubsidized], HPU [High Predator Unsubsidized] and HPS [High Predator Subsidized]) on Fernando de Noronha archipelago, investigating how invasive species presence and food supplementation affect population density, body size, and injury rates.

## Study Area Context

The study encompasses the following areas:

- **Main Island: 16.89 km<sup>2</sup>**
  - HPU: 11.424 km<sup>2</sup>
  - HPS: 5.466 km<sup>2</sup>
- **LPU: 1.33 km<sup>2</sup>**
- **TOTAL UNSUBSIDIZED (HPU + LPU): 12.754 km<sup>2</sup>**
- **TOTAL AREA (HPU + HPS + LPU): 18.22 km<sup>2</sup>**

## Package Setup

We'll use the `Require` package for better dependency management across all analyses.

```
if(!require("Require")){
  install.packages("Require")
}
library("Require")

Require::Require("data.table")
Require::Require("emmeans")
Require::Require("DHARMa")
Require::Require("glmmTMB")
Require::Require("performance")
Require::Require("RMark")
Require::Require("xlsx")
Require::Require("ggplot2")
```

```
Require::Require("multcompView")
Require::Require("dplyr")
Require::Require("pwr")
```

## Survey Effort Analysis

Before analyzing the biological data, we assessed whether our sampling effort was balanced across the study area.

### Loading Density Data

```
mabuia_dd <- fread("data/Density_Data.csv")
```

### Calculating Sampling Effort Distribution

```
pointsCounts <- mabuia_dd[, .N, by = "Site"]
pointsCounts[Site == "HPS", pointsArea := 5.466]
pointsCounts[Site == "HPU", pointsArea := 11.424]
pointsCounts[Site == "LPU", pointsArea := 1.33]
pointsCounts[, pointsDensity := N/pointsArea]
pointsCounts[, propPoints := N/(sum(pointsCounts[, N]))]
pointsCounts[, propSize := pointsArea/(sum(pointsCounts[, pointsArea]))]

print("Survey effort distribution:")
```

```
## [1] "Survey effort distribution:"
```

```
print(pointsCounts)
```

```
## Index: <Site>
##      Site      N pointsArea pointsDensity propPoints  propSize
##    <char> <int>      <num>      <num>      <num>      <num>
## 1:   HPS    58      5.466      10.611050  0.3866667  0.30000000
## 2:   HPU    55     11.424       4.814426  0.3666667  0.62700329
## 3:   LPU    37      1.330      27.819549  0.2466667  0.07299671
```

### Survey Effort Assessment

There is indeed a bit of unbalance but it is not as critical as it seems:

- **HPS:** 39% points over 30% area
- **HPU:** 37% points over 62% area
- **LPU:** 25% points over 7% area

The sampling is reasonably proportional to area, with only modest over-sampling of LPU and slight under-sampling of HPU relative to area.

## Population Density Analysis

### Step 1: Survey Area Calculation

Each survey point was a circle with 2m radius, giving us a standardized survey area.

```
mabuia_dd[, survey_area := pi * (2^2)]
print(paste("Survey area per point:", round(mabuia_dd$survey_area[1], 3), "m²"))
```

```
## [1] "Survey area per point: 12.566 m²"
```

The survey area is approximately 12.566 m<sup>2</sup> per point, which we'll use as an offset in our density models.

## Step 2: Creating Predictor Variables

We need to convert site identity into ecologically meaningful predictor variables based on site characteristics.

```
siteLevels <- factor(c("HPS", "HPU", "LPU"),
                     levels = c("HPS", "HPU", "LPU"))

# Invasive Species Presence (assuming key invasives like Cats and Tejus are largely absent from LPU)
mabuia_dd[Site == "HPS", InvasiveSpeciesPresence := 1]
mabuia_dd[Site == "HPU", InvasiveSpeciesPresence := 1]
mabuia_dd[Site == "LPU", InvasiveSpeciesPresence := 0]

# Food Supplementation
mabuia_dd[Site == "HPS", FoodSupplementation := 1]
mabuia_dd[Site == "HPU", FoodSupplementation := 0]
mabuia_dd[Site == "LPU", FoodSupplementation := 0]

# Convert to factors for modeling
mabuia_dd[, InvasiveSpeciesPresence := as.factor(InvasiveSpeciesPresence)]
mabuia_dd[, FoodSupplementation := as.factor(FoodSupplementation)]
head(mabuia_dd)
```

```
##      LocationID      Date    Time Location Site Island      Habitat Temp_oC
##      <char>      <char> <char> <char> <char> <char>      <char> <int>
## 1:      IPA56 2/11/2016 16:25 3 Paus  HPS  Main      Urbana      29
## 2:      IPA106 11/8/2016 8:16 Abreus  HPU  Main  Arbustiva Alta      30
## 3:      IPA107 11/8/2016 8:30 Abreus  HPU  Main  Arborea Baixa      30
## 4:      IPA108 11/8/2016 8:45 Abreus  HPU  Main  Gram\xednea      30
## 5:      IPA109 11/8/2016 9:07 Abreus  HPU  Main  Pedreira (costa)      30
## 6:      IPA110 11/8/2016 9:30 Abreus  HPU  Main  Pedreira (costa)      31
##      UTM      NOR      EAS Counts ObsMin survey_area InvasiveSpeciesPresence
##      <int> <int> <int> <int> <int>      <num>      <fctr>
## 1:      25 564037 9574144      3      7      12.56637      1
## 2:      25 564199 9573143      1      7      12.56637      1
## 3:      25 564452 9572968      1      7      12.56637      1
## 4:      25 564676 9572953      1      7      12.56637      1
## 5:      25 565228 9573056      0      7      12.56637      1
## 6:      25 565055 9573102      0      7      12.56637      1
##      FoodSupplementation
##      <fctr>
## 1:      1
## 2:      0
## 3:      0
## 4:      0
## 5:      0
## 6:      0
```

This creates a 2x2 factorial design where we can test the independent and interactive effects of invasive species presence and food supplementation.

### Step 3: Fitting the Main Density Model

We use a negative binomial GLM because count data often shows overdispersion (variance > mean).

```
model_hierarchical_main <- MASS::glm.nb(  
  Counts ~ InvasiveSpeciesPresence + FoodSupplementation + offset(log(survey_area)),  
  data = mabuia_dd,  
  na.action = na.omit  
)  
summary(model_hierarchical_main)
```

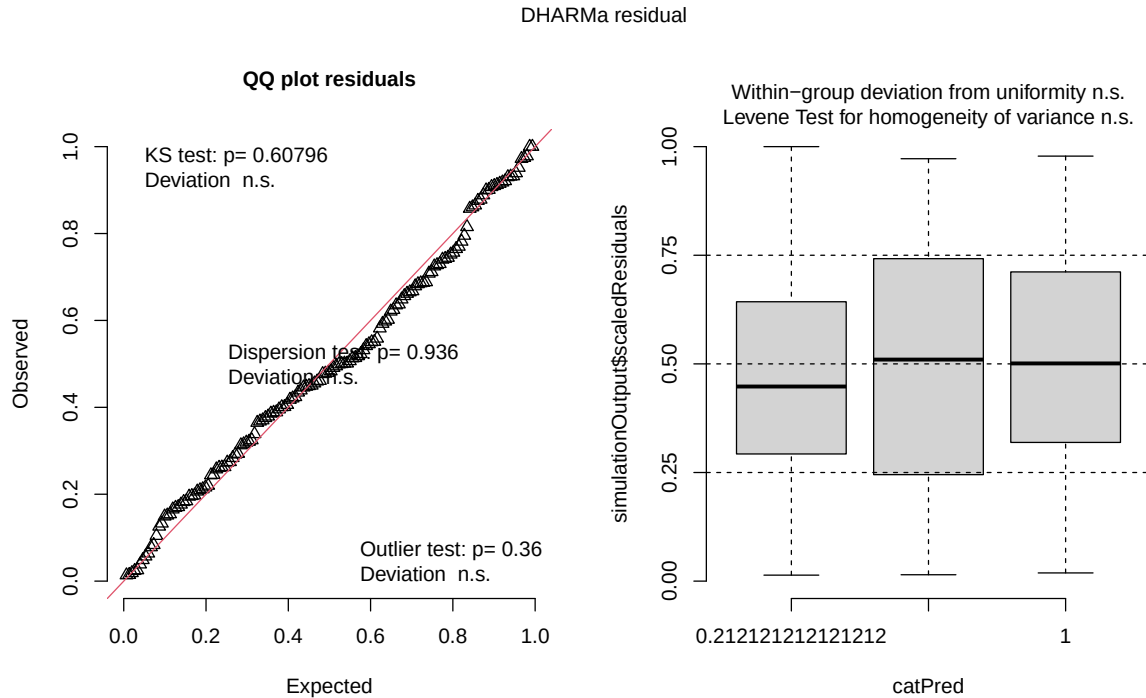
```
##  
## Call:  
## MASS::glm.nb(formula = Counts ~ InvasiveSpeciesPresence + FoodSupplementation +  
##   offset(log(survey_area)), data = mabuia_dd, na.action = na.omit,  
##   init.theta = 1.151932299, link = log)  
##  
## Coefficients:  
##  
##              Estimate Std. Error z value Pr(>|z|)  
## (Intercept)      -0.8897      0.1694  -5.252 1.51e-07 ***  
## InvasiveSpeciesPresence1 -0.9666      0.2318  -4.169 3.05e-05 ***  
## FoodSupplementation1      0.2687      0.2162   1.243  0.214  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## (Dispersion parameter for Negative Binomial(1.1519) family taken to be 1)  
##  
##    Null deviance: 184.46  on 149  degrees of freedom  
## Residual deviance: 165.06  on 147  degrees of freedom  
## AIC: 663.54  
##  
## Number of Fisher Scoring iterations: 1  
##  
##  
##              Theta:  1.152  
##             Std. Err.:  0.206  
##  
##  2 x log-likelihood:  -655.544
```

The offset term `log(survey_area)` allows us to model density (individuals per unit area) rather than raw counts.

### Step 4: Model Validation

Good model validation is critical for reliable inference.

```
r2(model_hierarchical_main)  
  
## # R2 for Generalized Linear Regression  
##   Nagelkerke's R2: 0.172  
  
sim_res <- simulateResiduals(model_hierarchical_main)  
plot(sim_res)
```



DHARMA provides robust residual diagnostics by simulating from the fitted model. Good residuals should show uniform distribution in QQ plots and no patterns in residual vs. fitted plots.

## Step 5: Alternative Model Formulations

We test several alternative model structures to ensure we've chosen the best approach.

### Interaction Model

```
model_interaction <- MASS::glm.nb(
  Counts ~ InvasiveSpeciesPresence * FoodSupplementation + offset(log(survey_area)),
  data = mabuia_dd
)
summary(model_interaction)
```

```
##
## Call:
## MASS::glm.nb(formula = Counts ~ InvasiveSpeciesPresence * FoodSupplementation +
##   offset(log(survey_area)), data = mabuia_dd, init.theta = 1.151932299,
##   link = log)
##
## Coefficients: (1 not defined because of singularities)
##
##               Estimate Std. Error z value
## (Intercept)    -0.8897    0.1694  -5.252
## InvasiveSpeciesPresence1 -0.9666    0.2318  -4.169
## FoodSupplementation1    0.2687    0.2162   1.243
## InvasiveSpeciesPresence1:FoodSupplementation1      NA         NA      NA
##
##               Pr(>|z|)
## (Intercept)  1.51e-07 ***
## InvasiveSpeciesPresence1  3.05e-05 ***
```

```
## FoodSupplementation1 0.214
## InvasiveSpeciesPresence1:FoodSupplementation1 NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Negative Binomial(1.1519) family taken to be 1)
##
##      Null deviance: 184.46  on 149  degrees of freedom
## Residual deviance: 165.06  on 147  degrees of freedom
## AIC: 663.54
##
## Number of Fisher Scoring iterations: 1
##
##
##      Theta:  1.152
##      Std. Err.:  0.206
##
## 2 x log-likelihood:  -655.544
```

## Mixed Effects Models

```
model_random <- glmmTMB::glmmTMB(
  Counts ~ InvasiveSpeciesPresence + FoodSupplementation + offset(log(survey_area)) + (1 | Site),
  family = nbinom2,
  data = mabuia_dd
)
summary(model_random)
```

```
## Family: nbinom2 ( log )
## Formula:
## Counts ~ InvasiveSpeciesPresence + FoodSupplementation + offset(log(survey_area)) +
##      (1 | Site)
## Data: mabuia_dd
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##      665.5    680.6   -327.8    655.5      145
##
## Random effects:
##
## Conditional model:
## Groups Name      Variance Std.Dev.
## Site (Intercept) 7.525e-10 2.743e-05
## Number of obs: 150, groups: Site, 3
##
## Dispersion parameter for nbinom2 family (): 1.15
##
## Conditional model:
##
##      Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -0.8897    0.1694  -5.252 1.51e-07 ***
## InvasiveSpeciesPresence1 -0.9666    0.2318  -4.169 3.05e-05 ***
## FoodSupplementation1    0.2687    0.2162   1.243  0.214
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```

model_random_interaction <- glmmTMB::glmmTMB(
  Counts ~ InvasiveSpeciesPresence * FoodSupplementation + offset(log(survey_area)) + (1 | Site),
  family = nbinom2,
  data = mabuia_dd
)
summary(model_random_interaction)

## Family: nbinom2 ( log )
## Formula:
## Counts ~ InvasiveSpeciesPresence * FoodSupplementation + offset(log(survey_area)) +
## (1 | Site)
## Data: mabuia_dd
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    665.5    680.6   -327.8    655.5      145
##
## Random effects:
##
## Conditional model:
## Groups Name      Variance Std.Dev.
## Site (Intercept) 7.525e-10 2.743e-05
## Number of obs: 150, groups: Site, 3
##
## Dispersion parameter for nbinom2 family (): 1.15
##
## Conditional model:
##
##              Estimate Std. Error z value
## (Intercept)    -0.8897    0.1694  -5.252
## InvasiveSpeciesPresence1    -0.9666    0.2318  -4.169
## FoodSupplementation1      0.2687    0.2162   1.243
## InvasiveSpeciesPresence1:FoodSupplementation1      NA      NA      NA
##              Pr(>|z|)
## (Intercept)    1.51e-07 ***
## InvasiveSpeciesPresence1    3.05e-05 ***
## FoodSupplementation1      0.214
## InvasiveSpeciesPresence1:FoodSupplementation1      NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

## Model Selection

```

model_comparison <- AIC(
  model_hierarchical_main,      # Original glm.nb model
  model_interaction,           # GLM with interaction
  model_random,                # GLMM with random effect
  model_random_interaction     # GLMM with both interaction + random
)
print("Model AIC Comparison:")

## [1] "Model AIC Comparison:"
print(model_comparison)

```

```
##              df      AIC
```



```
## model_hierarchical_main    4 663.5444
## model_interaction          4 663.5444
## model_random               5 665.5444
## model_random_interaction   5 665.5444
```

**Conclusion:** The original model has the best AIC, is interpretable, answers the ecological question clearly, and passes all residual diagnostics.

## Step 6: Estimated Marginal Means

```
emm_density_hierarchical <- emmeans(model_hierarchical_main,
                                     specs = ~ InvasiveSpeciesPresence + FoodSupplementation,
                                     type = "response",
                                     offset = 0)

summary(emm_density_hierarchical, infer = TRUE)
```

```
## InvasiveSpeciesPresence FoodSupplementation response      SE df asymp.LCL
## 0                        0                    0.411 0.0696 Inf      0.295
## 1                        0                    0.156 0.0247 Inf      0.115
## 0                        1                    0.537 0.1480 Inf      0.314
## 1                        1                    0.204 0.0301 Inf      0.153
## asymp.UCL null z.ratio p.value
##    0.573    1  -5.252 <0.0001
##    0.213    1 -11.730 <0.0001
##    0.921    1  -2.261  0.0238
##    0.273    1 -10.782 <0.0001
##
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
## Tests are performed on the log scale
```

Setting offset = 0 gives us density per unit area rather than total abundance.

## Step 7: Creating the Density Visualization

### Data Preparation for Plotting

```
density_df <- as.data.frame(summary(emm_density_hierarchical, infer = TRUE))

# Map factor combinations to meaningful site names
density_df$Site <- ifelse(
  density_df$InvasiveSpeciesPresence == 0 & density_df$FoodSupplementation == 1, "Hypothetical Site",
  ifelse(
    density_df$InvasiveSpeciesPresence == 0 & density_df$FoodSupplementation == 0, "LPU",
    ifelse(
      density_df$InvasiveSpeciesPresence == 1 & density_df$FoodSupplementation == 1, "HPS",
      "HPU"
    )
  )
)

density_df$Site <- factor(density_df$Site, levels = c("Hypothetical Site", "LPU", "HPS", "HPU"))

# Rename columns for clarity
```

```

names(density_df)[names(density_df) == 'response'] <- 'mean_density'
names(density_df)[names(density_df) == 'asympt.LCL'] <- 'lower_ci'
names(density_df)[names(density_df) == 'asympt.UCL'] <- 'upper_ci'
print(density_df)

```

```

## InvasiveSpeciesPresence FoodSupplementation mean_density SE df
## 0 0 0.4107918 0.06959005 Inf
## 1 0 0.1562612 0.02472831 Inf
## 0 1 0.5374264 0.14759153 Inf
## 1 1 0.2044318 0.03009994 Inf
## lower_ci upper_ci null z.ratio p.value Site
## 0.2947300 0.5725577 1 -5.252 <0.0001 LPU
## 0.1145905 0.2130854 1 -11.730 <0.0001 HPU
## 0.3137303 0.9206223 1 -2.261 0.0238 Hypothetical Site
## 0.1531863 0.2728204 1 -10.782 <0.0001 HPS
##
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
## Tests are performed on the log scale

```

### Statistical Significance Testing

```

pairs_emm <- contrast(emm_density_hierarchical, method = "pairwise", adjust = "tukey")
summary_pw <- summary(pairs_emm)
p_values <- summary_pw$p.value
print(p_values)

```

```

## [1] 0.0001790051 0.5993115525 0.0101326414 0.0079318154 0.5993115525
## [6] 0.0001790051

```

### Processing Significance Letters

```

# Helper function to clean emmeans factor level names
clean_level_name <- function(level_string) {
  cleaned <- gsub("InvasiveSpeciesPresence", "", level_string)
  cleaned <- gsub("FoodSupplementation", "", cleaned)
  cleaned <- gsub(" ", "", cleaned)
  return(cleaned)
}

contrast_names_raw <- as.character(summary_pw$contrast)
p_value_names <- sapply(contrast_names_raw, function(contrast) {
  parts <- strsplit(contrast, " / ")[[1]]
  key1 <- clean_level_name(trimws(parts[1]))
  key2 <- clean_level_name(trimws(parts[2]))
  return(paste(key1, key2, sep = "-"))
})
names(p_values) <- p_value_names

# Generate significance letters
letters_result <- multcompLetters(p_values)
significance_letters <- letters_result$Letters

# Add letters to plotting data

```

```

density_df$key <- paste0(density_df$InvasiveSpeciesPresence, density_df$FoodSupplementation)
density_df$significance_group <- significance_letters[density_df$key]
density_df$y_position_for_text <- density_df$upper_ci + 0.05

```

## Creating the Final Density Plot

```

two_line_labels <- c(
  "Hypothetical LPS\n(Low Predator / Subsidised)",
  "LPU\n(Low Predators/ Unsubsidised)",
  "HPS\n(High Predators/ Subsidised)",
  "HPU\n(High Predators/ Unsubsidised)"
)

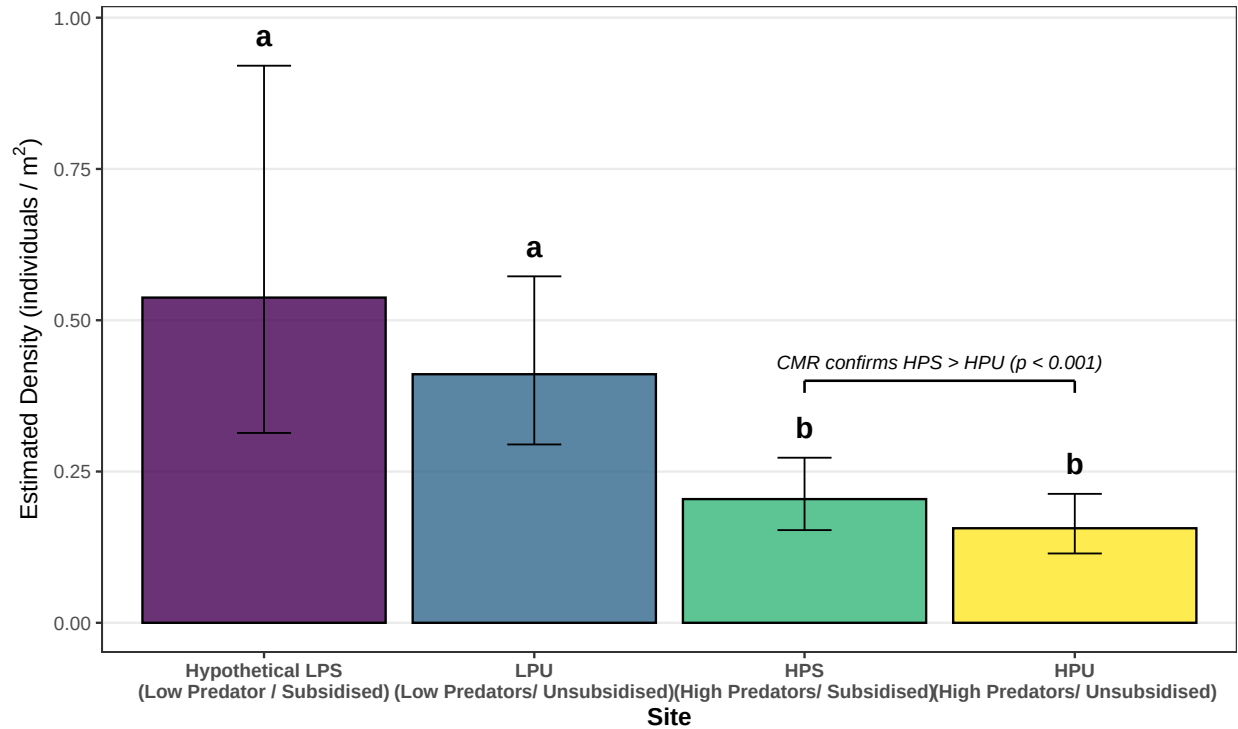
density_plot <- ggplot(density_df, aes(x = Site, y = mean_density, fill = Site)) +
  # Add bars and error bars from the GLM
  geom_bar(stat = "identity", color = "black", alpha = 0.8) +
  geom_errorbar(aes(ymin = lower_ci, ymax = upper_ci),
    width = 0.2, color = "black", linewidth = 0.5) +
  # Add significance letters from the GLM
  geom_text(aes(y = y_position_for_text, label = significance_group),
    size = 6, fontface = "bold", color = "black") +

  # --- ANNOTATION FOR CMR RESULT ---
  # Add a bracket connecting HPS and HPU bars
  annotate("segment", x = "HPS", xend = "HPU", y = 0.40, yend = 0.40,
    color = "black", linewidth = 0.7) +
  # New position for the left vertical tick (pointing down).
  annotate("segment", x = "HPS", xend = "HPS", y = 0.40, yend = 0.38,
    color = "black", linewidth = 0.7) +
  # New position for the right vertical tick (pointing down).
  annotate("segment", x = "HPU", xend = "HPU", y = 0.40, yend = 0.38,
    color = "black", linewidth = 0.7) +
  # New position for the text, slightly above the bracket line.
  annotate("text", x = 3.5, y = 0.43, label = "CMR confirms HPS > HPU (p < 0.001)",
    fontface = "italic", color = "black", size = 4) +

  # --- SCALES, LABELS, and THEMING ---
  scale_x_discrete(labels = two_line_labels) +
  scale_fill_viridis_d(option = "D") +
  labs(
    y = expression("Estimated Density (individuals / m^2)")
  ) +
  theme_bw(base_size = 14) +
  theme(
    axis.title = element_text(face = "bold"),
    axis.text.x = element_text(face = "bold"),
    panel.grid.major.x = element_blank(),
    panel.grid.minor = element_blank(),
    legend.position = "none"
  )

print(density_plot)

```



## Mark-Recapture Analysis

Analysis was performed using the Poisson-log normal mark-resight model to separate detection probability from true abundance.

### Step 1: Loading CMR Data

```
ch <- read.xlsx(file = file.path(getwd(), "data/CRM_Data.xlsx"),
               header = TRUE,
               sheetName = "ch",
               as.data.frame = TRUE)
ch$ch <- as.character(ch$ch)
ch$colonies <- as.factor(ch$colonies)

covars <- read.xlsx(file = file.path(getwd(), "data/CRM_Data.xlsx"),
                   header = TRUE,
                   sheetName = "covars",
                   as.data.frame = TRUE)
```

### Step 2: Creating Model Variables

```
nocc <- 6 # Number of occasions
groupsNames <- unique(covars$colonies)
nGroups <- length(groupsNames)

# Create matrices for different count types
unmarkedSeen <- matrix(covars[, "unmarkedSeen"],
                      nrow = nGroups,
```

```

        ncol = nocc,
        byrow = TRUE)

markedUnidentified <- matrix(covars[, "markedUnidentified"],
                             nrow = nGroups,
                             ncol = nocc,
                             byrow = TRUE)

knownMarks <- matrix(covars[, "knownMarks"],
                     nrow = nGroups,
                     ncol = nocc,
                     byrow = TRUE)

effort <- matrix(covars[, "effort"],
                 nrow = nGroups,
                 ncol = nocc,
                 byrow = TRUE)

site <- matrix(covars[, "site"],
               nrow = nGroups,
               ncol = nocc,
               byrow = TRUE)

```

### Step 3: Data Processing

```

mabuya.process <- process.data(ch,
                              model = "PoissonMR",
                              time.intervals = rep(1, times = (nocc-1)),
                              groups = "colonies",
                              counts = list("Unmarked Seen" = unmarkedSeen,
                                             "Marked Unidentified" = markedUnidentified,
                                             "Known Marks" = knownMarks))

mabuya.ddl <- make.design.data(mabuya.process)

```

### Step 4: Adding Covariates

Now we include covariates such as effort and site type where they should be placed.

#### For Alpha (Detection Probability)

```

mabuya.ddl$alpha$effort <- as.numeric(apply(X = mabuya.ddl$alpha, 1, function(x){
  col <- x[["group"]]
  occ <- as.numeric(x[["time"]])
  eff <- covars[covars$colonies==col&covars$occasion==occ, "effort"]
}))

mabuya.ddl$alpha$site <- as.character(apply(X = mabuya.ddl$alpha, 1, function(x){
  col <- x[["group"]]
  occ <- as.numeric(x[["time"]])
  site <- covars[covars$colonies==col&covars$occasion==occ, "site"]
}))

```

## For U (Abundance)

```
mabuya.ddl$U$site <- as.character(apply(X = mabuya.ddl$U, 1, function(x){
  col <- x[["group"]]
  occ <- as.numeric(x[["time"]])
  site <- covars[covars$colonies==col&covars$occasion==occ, "site"]
})))
```

## Step 5: Defining Model Formulas

```
# Parameter formulas for alpha (detection probability)
alpha.dot <- list(formula = ~ 1)
alpha.effort <- list(formula = ~ effort)
alpha.site <- list(formula = ~ site)
alpha.additive <- list(formula = ~ effort + site)
alpha.interaction <- list(formula = ~ effort * site)

# Parameters that will be the same in all models
U.site <- list(formula = ~ site)
Phi.fixed1 <- list(formula = ~ 1, fixed = 1)
zero <- list(formula = ~ 1, fixed = 0) # For sigma and Gammas
```

## Step 6: Building Model List

```
model.list <- list(
  "alpha(.) U(site)" = list(
    alpha = alpha.dot,
    U = U.site,
    Phi = Phi.fixed1,
    sigma = zero,
    GammaDoublePrime = zero,
    GammaPrime = zero
  ),
  "alpha(effort) U(site)" = list(
    alpha = alpha.effort,
    U = U.site,
    Phi = Phi.fixed1,
    sigma = zero,
    GammaDoublePrime = zero,
    GammaPrime = zero
  ),
  "alpha(site) U(site)" = list(
    alpha = alpha.site,
    U = U.site,
    Phi = Phi.fixed1,
    sigma = zero,
    GammaDoublePrime = zero,
    GammaPrime = zero
  ),
  "alpha(effort+site) U(site)" = list(
    alpha = alpha.additive,
    U = U.site,
    Phi = Phi.fixed1,
```

```

    sigma = zero,
    GammaDoublePrime = zero,
    GammaPrime = zero
  ),
  "alpha(effort*site) U(site)" = list(
    alpha = alpha.interaction,
    U = U.site,
    Phi = Phi.fixed1,
    sigma = zero,
    GammaDoublePrime = zero,
    GammaPrime = zero
  )
)

```

## Step 7: Fitting All Models

```

mabuya_model_1 <- mark(
  data = mabuya.process,
  ddl = mabuya.ddl,
  model.parameters = model.list[["alpha(.) U(site)"]],
  delete = TRUE
)

```

```

##
## Output summary for PoissonMR model
## Name : alpha(~1)sigma(~1)U(~site)Phi(~1)Gamma' (~1)Gamma' (~1)
##
## Npar : 3
## -2lnL: 872.2775
## AICc : 878.3273
##
## Beta
##
## estimate se lcl ucl
## alpha:(Intercept) -1.354831 0.0940721 -1.539212 -1.170450
## U:(Intercept) 4.158624 0.1063001 3.950276 4.366973
## U:siteHPU -1.760433 0.1292842 -2.013830 -1.507036
##
##
## Real Parameter alpha
##
## 1 2 3 4 5
## Group:coloniesAmericano 0.2579909 0.2579909 0.2579909 0.2579909 0.2579909
## Group:coloniesCapimAcu 0.2579909 0.2579909 0.2579909 0.2579909 0.2579909
## Group:coloniesForteBoldro 0.2579909 0.2579909 0.2579909 0.2579909 0.2579909
## Group:coloniesLeao 0.2579909 0.2579909 0.2579909 0.2579909 0.2579909
## Group:coloniesPedreiraSueste 0.2579909 0.2579909 0.2579909 0.2579909 0.2579909
## Group:coloniesPiquinho 0.2579909 0.2579909 0.2579909 0.2579909 0.2579909
## Group:coloniesPraiaBoldro 0.2579909 0.2579909 0.2579909 0.2579909 0.2579909
## Group:coloniesTejuAcu 0.2579909 0.2579909 0.2579909 0.2579909 0.2579909
##
## 6
## Group:coloniesAmericano 0.2579909
## Group:coloniesCapimAcu 0.2579909
## Group:coloniesForteBoldro 0.2579909
## Group:coloniesLeao 0.2579909

```

```

## Group:coloniesPedreiraSueste 0.2579909
## Group:coloniesPiquinho      0.2579909
## Group:coloniesPraiaBoldro   0.2579909
## Group:coloniesTejuAcu       0.2579909
##
##
## Real Parameter sigma
##
##      1  2  3  4  5  6
## Group:coloniesAmericano      NA NA NA NA NA NA
## Group:coloniesCapimAcu       NA NA NA NA NA NA
## Group:coloniesForteBoldro    NA NA NA NA NA NA
## Group:coloniesLeao           NA NA NA NA NA NA
## Group:coloniesPedreiraSueste NA NA NA NA NA NA
## Group:coloniesPiquinho       NA NA NA NA NA NA
## Group:coloniesPraiaBoldro    NA NA NA NA NA NA
## Group:coloniesTejuAcu        NA NA NA NA NA NA
##
##
## Real Parameter U
##
##      1      2      3      4      5
## Group:coloniesAmericano      63.98345 63.98345 63.98345 63.98345 63.98345
## Group:coloniesCapimAcu       11.00326 11.00326 11.00326 11.00326 11.00326
## Group:coloniesForteBoldro    63.98345 63.98345 63.98345 63.98345 63.98345
## Group:coloniesLeao           11.00326 11.00326 11.00326 11.00326 11.00326
## Group:coloniesPedreiraSueste 11.00326 11.00326 11.00326 11.00326 11.00326
## Group:coloniesPiquinho       11.00326 11.00326 11.00326 11.00326 11.00326
## Group:coloniesPraiaBoldro    63.98345 63.98345 63.98345 63.98345 63.98345
## Group:coloniesTejuAcu        63.98345 63.98345 63.98345 63.98345 63.98345
##
##      6
## Group:coloniesAmericano      63.98345
## Group:coloniesCapimAcu       11.00326
## Group:coloniesForteBoldro    63.98345
## Group:coloniesLeao           11.00326
## Group:coloniesPedreiraSueste 11.00326
## Group:coloniesPiquinho       11.00326
## Group:coloniesPraiaBoldro    63.98345
## Group:coloniesTejuAcu        63.98345
##
##
## Real Parameter Phi
## Group:coloniesAmericano
##      1  2  3  4  5
##      1
##      2
##      3
##      4
##      5
##
## Group:coloniesCapimAcu
##      1  2  3  4  5
##      1
##      2
##      3
##      4

```



```

## 5
##
## Group:coloniesForteBoldro
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesLeao
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPedreiraSueste
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPiquinho
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPraiaBoldro
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesTejuAcu
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
##
## Real Parameter GammaDoublePrime
## Group:coloniesAmericano
## 1 2 3 4 5

```

```

## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesCapimAcu
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesForteBoldro
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesLeao
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPedreiraSueste
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPiquinho
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPraiaBoldro
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##

```

```

## Group:coloniesTejuAcu
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
##
## Real Parameter GammaPrime
## Group:coloniesAmericano
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesCapimAcu
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesForteBoldro
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesLeao
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesPedreiraSueste
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesPiquinho
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesPraiaBoldro
## 2 3 4 5

```

```

## 1
## 2
## 3
## 4
##
## Group:coloniesTejuAcu
##   2 3 4 5
## 1
## 2
## 3
## 4

mabuya_model_2 <- mark(
  data = mabuya.process,
  ddl = mabuya.ddl,
  model.parameters = model.list[["alpha(effort) U(site)"]],
  delete = TRUE
)

##
## Output summary for PoissonMR model
## Name : alpha(~effort)sigma(~1)U(~site)Phi(~1)Gamma'(~1)Gamma'(~1)
##
## Npar : 4
## -2lnL: 849.1093
## AICc : 857.1925
##
## Beta
##           estimate      se      lcl      ucl
## alpha:(Intercept) -1.8070139 0.1359359 -2.0734483 -1.5405795
## alpha:effort      0.0189359 0.0039292  0.0112346  0.0266371
## U:(Intercept)     4.1289170 0.1067975  3.9195938  4.3382401
## U:siteHPU         -1.5993993 0.1343945 -1.8628125 -1.3359862
##
##
## Real Parameter alpha
##           1           2           3           4           5
## Group:coloniesAmericano 0.4230688 0.2180620 0.2635223 0.2397169 0.2180620
## Group:coloniesCapimAcu  0.2442994 0.2021552 0.2180620 0.2180620 0.2180620
## Group:coloniesForteBoldro 0.3184600 0.2180620 0.2635223 0.2635223 0.3008733
## Group:coloniesLeao      0.2308082 0.2397169 0.2099580 0.2264788 0.2099580
## Group:coloniesPedreiraSueste 0.2397169 0.2264788 0.2442994 0.2308082 0.2222305
## Group:coloniesPiquinho  0.2308082 0.2060197 0.2736938 0.2180620 0.2180620
## Group:coloniesPraiaBoldro 0.2308082 0.3008733 0.3500851 0.2789258 0.2060197
## Group:coloniesTejuAcu   0.2180620 0.1909914 0.1909914 0.1983633 0.1983633
##           6
## Group:coloniesAmericano 0.2180620
## Group:coloniesCapimAcu  0.2352204
## Group:coloniesForteBoldro 0.2635223
## Group:coloniesLeao      0.2180620
## Group:coloniesPedreiraSueste 0.2139716
## Group:coloniesPiquinho  0.1983633
## Group:coloniesPraiaBoldro 0.3500851
## Group:coloniesTejuAcu   0.1983633
##

```

```

##
## Real Parameter sigma
##           1  2  3  4  5  6
## Group:coloniesAmericano      NA NA NA NA NA NA
## Group:coloniesCapimAcu       NA NA NA NA NA NA
## Group:coloniesForteBoldro    NA NA NA NA NA NA
## Group:coloniesLeao           NA NA NA NA NA NA
## Group:coloniesPedreiraSueste NA NA NA NA NA NA
## Group:coloniesPiquinho       NA NA NA NA NA NA
## Group:coloniesPraiaBoldro    NA NA NA NA NA NA
## Group:coloniesTejuAcu        NA NA NA NA NA NA
##
##
## Real Parameter U
##           1           2           3           4           5
## Group:coloniesAmericano      62.11062 62.11062 62.11062 62.11062 62.11062
## Group:coloniesCapimAcu       12.54745 12.54745 12.54745 12.54745 12.54745
## Group:coloniesForteBoldro    62.11062 62.11062 62.11062 62.11062 62.11062
## Group:coloniesLeao           12.54745 12.54745 12.54745 12.54745 12.54745
## Group:coloniesPedreiraSueste 12.54745 12.54745 12.54745 12.54745 12.54745
## Group:coloniesPiquinho       12.54745 12.54745 12.54745 12.54745 12.54745
## Group:coloniesPraiaBoldro    62.11062 62.11062 62.11062 62.11062 62.11062
## Group:coloniesTejuAcu        62.11062 62.11062 62.11062 62.11062 62.11062
##           6
## Group:coloniesAmericano      62.11062
## Group:coloniesCapimAcu       12.54745
## Group:coloniesForteBoldro    62.11062
## Group:coloniesLeao           12.54745
## Group:coloniesPedreiraSueste 12.54745
## Group:coloniesPiquinho       12.54745
## Group:coloniesPraiaBoldro    62.11062
## Group:coloniesTejuAcu        62.11062
##
##
## Real Parameter Phi
## Group:coloniesAmericano
##   1  2  3  4  5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesCapimAcu
##   1  2  3  4  5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesForteBoldro
##   1  2  3  4  5
## 1

```

```

## 2
## 3
## 4
## 5
##
## Group:coloniesLeao
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPedreiraSueste
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPiquinho
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPraiaBoldro
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesTejuAcu
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
##
## Real Parameter GammaDoublePrime
## Group:coloniesAmericano
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5

```

```

##
## Group:coloniesCapimAcu
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesForteBoldro
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesLeao
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPedreiraSueste
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPiquinho
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPraiaBoldro
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesTejuAcu
## 1 2 3 4 5
## 1
## 2
## 3

```

```

## 4
## 5
##
##
## Real Parameter GammaPrime
## Group:coloniesAmericano
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesCapimAcu
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesForteBoldro
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesLeao
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesPedreiraSueste
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesPiquinho
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesPraiaBoldro
## 2 3 4 5
## 1
## 2
## 3
## 4
##

```



```
## Group:coloniesTejuAcu
## 2 3 4 5
## 1
## 2
## 3
## 4
```

```
mabuya_model_3 <- mark(
  data = mabuya.process,
  ddl = mabuya.ddl,
  model.parameters = model.list[["alpha(site) U(site)"]],
  delete = TRUE
)
```

```
##
## Output summary for PoissonMR model
## Name : alpha(~site)sigma(~1)U(~site)Phi(~1)Gamma'(~1)Gamma'(~1)
##
## Npar : 4
## -2lnL: 871.1816
## AICc : 879.2648
##
```

```
## Beta
##           estimate      se      lcl      ucl
## alpha:(Intercept) -1.4182458 0.1139607 -1.6416087 -1.1948830
## alpha:siteHPU      0.2142728 0.2019031 -0.1814573  0.6100029
## U:(Intercept)      4.2220392 0.1242465  3.9785161  4.4655623
## U:siteHPU          -1.9747053 0.2397484 -2.4446121 -1.5047985
##
```

```
## Real Parameter alpha
```

|                                 | 1         | 2         | 3         | 4         | 5         |
|---------------------------------|-----------|-----------|-----------|-----------|-----------|
| ## Group:coloniesAmericano      | 0.2421384 | 0.2421384 | 0.2421384 | 0.2421384 | 0.2421384 |
| ## Group:coloniesCapimAcu       | 0.2999999 | 0.2999999 | 0.2999999 | 0.2999999 | 0.2999999 |
| ## Group:coloniesForteBoldro    | 0.2421384 | 0.2421384 | 0.2421384 | 0.2421384 | 0.2421384 |
| ## Group:coloniesLeao           | 0.2999999 | 0.2999999 | 0.2999999 | 0.2999999 | 0.2999999 |
| ## Group:coloniesPedreiraSueste | 0.2999999 | 0.2999999 | 0.2999999 | 0.2999999 | 0.2999999 |
| ## Group:coloniesPiquinho       | 0.2999999 | 0.2999999 | 0.2999999 | 0.2999999 | 0.2999999 |
| ## Group:coloniesPraiaBoldro    | 0.2421384 | 0.2421384 | 0.2421384 | 0.2421384 | 0.2421384 |
| ## Group:coloniesTejuAcu        | 0.2421384 | 0.2421384 | 0.2421384 | 0.2421384 | 0.2421384 |

|                                 | 6         |
|---------------------------------|-----------|
| ## Group:coloniesAmericano      | 0.2421384 |
| ## Group:coloniesCapimAcu       | 0.2999999 |
| ## Group:coloniesForteBoldro    | 0.2421384 |
| ## Group:coloniesLeao           | 0.2999999 |
| ## Group:coloniesPedreiraSueste | 0.2999999 |
| ## Group:coloniesPiquinho       | 0.2999999 |
| ## Group:coloniesPraiaBoldro    | 0.2421384 |
| ## Group:coloniesTejuAcu        | 0.2421384 |

```
## Real Parameter sigma
```

|                            | 1  | 2  | 3  | 4  | 5  | 6  |
|----------------------------|----|----|----|----|----|----|
| ## Group:coloniesAmericano | NA | NA | NA | NA | NA | NA |
| ## Group:coloniesCapimAcu  | NA | NA | NA | NA | NA | NA |

```

## Group:coloniesForteBoldro      NA NA NA NA NA NA
## Group:coloniesLeao             NA NA NA NA NA NA
## Group:coloniesPedreiraSueste   NA NA NA NA NA NA
## Group:coloniesPiquinho         NA NA NA NA NA NA
## Group:coloniesPraiaBoldro      NA NA NA NA NA NA
## Group:coloniesTejuAcu          NA NA NA NA NA NA
##
##
## Real Parameter U
##              1              2              3              4              5
## Group:coloniesAmericano        68.172360 68.172360 68.172360 68.172360 68.172360
## Group:coloniesCapimAcu          9.462474 9.462474 9.462474 9.462474 9.462474
## Group:coloniesForteBoldro      68.172360 68.172360 68.172360 68.172360 68.172360
## Group:coloniesLeao             9.462474 9.462474 9.462474 9.462474 9.462474
## Group:coloniesPedreiraSueste   9.462474 9.462474 9.462474 9.462474 9.462474
## Group:coloniesPiquinho         9.462474 9.462474 9.462474 9.462474 9.462474
## Group:coloniesPraiaBoldro      68.172360 68.172360 68.172360 68.172360 68.172360
## Group:coloniesTejuAcu          68.172360 68.172360 68.172360 68.172360 68.172360
##              6
## Group:coloniesAmericano        68.172360
## Group:coloniesCapimAcu          9.462474
## Group:coloniesForteBoldro      68.172360
## Group:coloniesLeao             9.462474
## Group:coloniesPedreiraSueste   9.462474
## Group:coloniesPiquinho         9.462474
## Group:coloniesPraiaBoldro      68.172360
## Group:coloniesTejuAcu          68.172360
##
##
## Real Parameter Phi
## Group:coloniesAmericano
##   1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesCapimAcu
##   1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesForteBoldro
##   1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##

```

```

## Group:coloniesLeao
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPedreiraSueste
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPiquinho
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPraiaBoldro
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesTejuAcu
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
##
## Real Parameter GammaDoublePrime
## Group:coloniesAmericano
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesCapimAcu
## 1 2 3 4 5
## 1
## 2

```

```

## 3
## 4
## 5
##
## Group:coloniesForteBoldro
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesLeao
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPedreiraSueste
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPiquinho
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPraiaBoldro
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesTejuAcu
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
##
## Real Parameter GammaPrime

```

```

## Group:coloniesAmericano
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesCapimAcu
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesForteBoldro
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesLeao
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesPedreiraSueste
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesPiquinho
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesPraiaBoldro
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesTejuAcu
## 2 3 4 5
## 1
## 2
## 3

```

```
## 4
mabuya_model_4 <- mark(
  data = mabuya.process,
  ddl = mabuya.ddl,
  model.parameters = model.list[["alpha(effort+site) U(site)"]],
  delete = TRUE
)

##
## Output summary for PoissonMR model
## Name : alpha(~effort + site)sigma(~1)U(~site)Phi(~1)Gamma'(~1)Gamma'(~1)
##
## Npar : 5
## -2lnL: 845.2658
## AICc : 855.3908
##
## Beta
##          estimate      se      lcl      ucl
## alpha:(Intercept) -1.9612543 0.1597802 -2.2744234 -1.6480852
## alpha:effort      0.0205160 0.0040357  0.0126060  0.0284260
## alpha:siteHPU     0.4154279 0.2066882  0.0103191  0.8205367
## U:(Intercept)     4.2427291 0.1249664  3.9977950  4.4876632
## U:siteHPU         -2.0009952 0.2402727 -2.4719298 -1.5300606
##
##
## Real Parameter alpha
##          1          2          3          4          5
## Group:coloniesAmericano 0.3924079 0.1913762 0.2349567 0.2120498 0.1913762
## Group:coloniesCapimAcu 0.3279186 0.2670952 0.2899386 0.2899386 0.2899386
## Group:coloniesForteBoldro 0.2884615 0.1913762 0.2349567 0.2349567 0.2712426
## Group:coloniesLeao 0.3083444 0.3212595 0.2782826 0.3020828 0.2782826
## Group:coloniesPedreiraSueste 0.3212595 0.3020828 0.3279186 0.3083444 0.2959484
## Group:coloniesPiquinho 0.3083444 0.2726315 0.3708736 0.2899386 0.2899386
## Group:coloniesPraiaBoldro 0.2035251 0.2712426 0.3196229 0.2498722 0.1799525
## Group:coloniesTejuAcu 0.1913762 0.1657746 0.1657746 0.1727182 0.1727182
##          6
## Group:coloniesAmericano 0.1913762
## Group:coloniesCapimAcu 0.3147357
## Group:coloniesForteBoldro 0.2349567
## Group:coloniesLeao 0.2899386
## Group:coloniesPedreiraSueste 0.2840508
## Group:coloniesPiquinho 0.2616713
## Group:coloniesPraiaBoldro 0.3196229
## Group:coloniesTejuAcu 0.1727182
##
##
## Real Parameter sigma
##          1  2  3  4  5  6
## Group:coloniesAmericano NA NA NA NA NA NA
## Group:coloniesCapimAcu NA NA NA NA NA NA
## Group:coloniesForteBoldro NA NA NA NA NA NA
## Group:coloniesLeao NA NA NA NA NA NA
## Group:coloniesPedreiraSueste NA NA NA NA NA NA
## Group:coloniesPiquinho NA NA NA NA NA NA
```

```

## Group:coloniesPraiaBoldro    NA NA NA NA NA NA
## Group:coloniesTejuAcu        NA NA NA NA NA NA
##
##
## Real Parameter U
##           1           2           3           4           5
## Group:coloniesAmericano      69.597532 69.597532 69.597532 69.597532 69.597532
## Group:coloniesCapimAcu        9.409632  9.409632  9.409632  9.409632  9.409632
## Group:coloniesForteBoldro     69.597532 69.597532 69.597532 69.597532 69.597532
## Group:coloniesLeao            9.409632  9.409632  9.409632  9.409632  9.409632
## Group:coloniesPedreiraSueste  9.409632  9.409632  9.409632  9.409632  9.409632
## Group:coloniesPiquinho        9.409632  9.409632  9.409632  9.409632  9.409632
## Group:coloniesPraiaBoldro     69.597532 69.597532 69.597532 69.597532 69.597532
## Group:coloniesTejuAcu         69.597532 69.597532 69.597532 69.597532 69.597532
##           6
## Group:coloniesAmericano      69.597532
## Group:coloniesCapimAcu        9.409632
## Group:coloniesForteBoldro     69.597532
## Group:coloniesLeao            9.409632
## Group:coloniesPedreiraSueste  9.409632
## Group:coloniesPiquinho        9.409632
## Group:coloniesPraiaBoldro     69.597532
## Group:coloniesTejuAcu         69.597532
##
##
## Real Parameter Phi
## Group:coloniesAmericano
##   1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesCapimAcu
##   1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesForteBoldro
##   1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesLeao
##   1 2 3 4 5
## 1
## 2

```

```

## 3
## 4
## 5
##
## Group:coloniesPedreiraSueste
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPiquinho
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPraiaBoldro
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesTejuAcu
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
##
## Real Parameter GammaDoublePrime
## Group:coloniesAmericano
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesCapimAcu
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##

```



```

## Group:coloniesForteBoldro
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesLeao
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPedreiraSueste
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPiquinho
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPraiaBoldro
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesTejuAcu
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
##
## Real Parameter GammaPrime
## Group:coloniesAmericano
## 2 3 4 5
## 1
## 2

```

```

## 3
## 4
##
## Group:coloniesCapimAcu
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesForteBoldro
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesLeao
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesPedreiraSueste
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesPiquinho
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesPraiaBoldro
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesTejuAcu
## 2 3 4 5
## 1
## 2
## 3
## 4
mabuya_model_5 <- mark(
  data = mabuya.process,

```

```

ddl = mabuya.ddl,
model.parameters = model.list[["alpha(effort*site) U(site)"]],
delete = TRUE
)

##
## Output summary for PoissonMR model
## Name : alpha(~effort * site)sigma(~1)U(~site)Phi(~1)Gamma'(~1)Gamma'(~1)
##
## Npar : 6
## -2lnL: 845.0471
## AICc : 857.2225
##
## Beta
##
## estimate      se      lcl      ucl
## alpha:(Intercept) -1.9710750 0.1612669 -2.2871581 -1.6549919
## alpha:effort      0.0208696 0.0041051 0.0128237 0.0289156
## alpha:siteHPU      0.5931867 0.4306514 -0.2508901 1.4372636
## alpha:effort:siteHPU -0.0103921 0.0221580 -0.0538218 0.0330375
## U:(Intercept)      4.2434972 0.1249911 3.9985146 4.4884797
## U:siteHPU          -2.0002849 0.2403150 -2.4713023 -1.5292675
##
##
## Real Parameter alpha
##
## 1 2 3 4 5
## Group:coloniesAmericano 0.3955042 0.1905138 0.2347265 0.2114678 0.1905138
## Group:coloniesCapimAcu 0.3141575 0.2829074 0.2950160 0.2950160 0.2950160
## Group:coloniesForteBoldro 0.2891996 0.1905138 0.2347265 0.2347265 0.2716483
## Group:coloniesLeao 0.3044364 0.3108831 0.2888983 0.3012633 0.2888983
## Group:coloniesPedreiraSueste 0.3108831 0.3012633 0.3141575 0.3044364 0.2981233
## Group:coloniesPiquinho 0.3044364 0.2858871 0.3345411 0.2950160 0.2950160
## Group:coloniesPraiaBoldro 0.2028230 0.2716483 0.3210077 0.2498922 0.1789517
## Group:coloniesTejuAcu 0.1905138 0.1646196 0.1646196 0.1716361 0.1716361
##
## 6
## Group:coloniesAmericano 0.1905138
## Group:coloniesCapimAcu 0.3076429
## Group:coloniesForteBoldro 0.2347265
## Group:coloniesLeao 0.2950160
## Group:coloniesPedreiraSueste 0.2919411
## Group:coloniesPiquinho 0.2799587
## Group:coloniesPraiaBoldro 0.3210077
## Group:coloniesTejuAcu 0.1716361
##
##
## Real Parameter sigma
##
## 1 2 3 4 5 6
## Group:coloniesAmericano NA NA NA NA NA NA
## Group:coloniesCapimAcu NA NA NA NA NA NA
## Group:coloniesForteBoldro NA NA NA NA NA NA
## Group:coloniesLeao NA NA NA NA NA NA
## Group:coloniesPedreiraSueste NA NA NA NA NA NA
## Group:coloniesPiquinho NA NA NA NA NA NA
## Group:coloniesPraiaBoldro NA NA NA NA NA NA
## Group:coloniesTejuAcu NA NA NA NA NA NA

```

```

##
##
## Real Parameter U
##          1          2          3          4          5
## Group:coloniesAmericano 69.651008 69.651008 69.651008 69.651008 69.651008
## Group:coloniesCapimAcu  9.423554 9.423554 9.423554 9.423554 9.423554
## Group:coloniesForteBoldro 69.651008 69.651008 69.651008 69.651008 69.651008
## Group:coloniesLeao      9.423554 9.423554 9.423554 9.423554 9.423554
## Group:coloniesPedreiraSueste 9.423554 9.423554 9.423554 9.423554 9.423554
## Group:coloniesPiquinho  9.423554 9.423554 9.423554 9.423554 9.423554
## Group:coloniesPraiaBoldro 69.651008 69.651008 69.651008 69.651008 69.651008
## Group:coloniesTejuAcu   69.651008 69.651008 69.651008 69.651008 69.651008
##          6
## Group:coloniesAmericano 69.651008
## Group:coloniesCapimAcu  9.423554
## Group:coloniesForteBoldro 69.651008
## Group:coloniesLeao      9.423554
## Group:coloniesPedreiraSueste 9.423554
## Group:coloniesPiquinho  9.423554
## Group:coloniesPraiaBoldro 69.651008
## Group:coloniesTejuAcu   69.651008
##
##
## Real Parameter Phi
## Group:coloniesAmericano
##   1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesCapimAcu
##   1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesForteBoldro
##   1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesLeao
##   1 2 3 4 5
## 1
## 2
## 3
## 4

```

```

## 5
##
## Group:coloniesPedreiraSueste
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPiquinho
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPraiaBoldro
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesTejuAcu
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
##
## Real Parameter GammaDoublePrime
## Group:coloniesAmericano
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesCapimAcu
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesForteBoldro
## 1 2 3 4 5

```

```

## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesLeao
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPedreiraSueste
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPiquinho
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesPraiaBoldro
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
## Group:coloniesTejuAcu
## 1 2 3 4 5
## 1
## 2
## 3
## 4
## 5
##
##
## Real Parameter GammaPrime
## Group:coloniesAmericano
## 2 3 4 5
## 1
## 2
## 3
## 4

```

```

##
## Group:coloniesCapimAcu
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesForteBoldro
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesLeao
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesPedreiraSueste
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesPiquinho
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesPraiaBoldro
## 2 3 4 5
## 1
## 2
## 3
## 4
##
## Group:coloniesTejuAcu
## 2 3 4 5
## 1
## 2
## 3
## 4

```

## Step 8: Model Selection and Best Model Results

```
tb <- collect.models(lx = c("mabuya_model_1", "mabuya_model_2",
                           "mabuya_model_3", "mabuya_model_4",
                           "mabuya_model_5"))

tbDT <- data.table(print(tb))

##                                model npar
## 4 alpha(~effort + site)sigma(~1)U(~site)Phi(~1)Gamma' (~1)Gamma' (~1)    5
## 2      alpha(~effort)sigma(~1)U(~site)Phi(~1)Gamma' (~1)Gamma' (~1)    4
## 5 alpha(~effort * site)sigma(~1)U(~site)Phi(~1)Gamma' (~1)Gamma' (~1)    6
## 1      alpha(~1)sigma(~1)U(~site)Phi(~1)Gamma' (~1)Gamma' (~1)    3
## 3      alpha(~site)sigma(~1)U(~site)Phi(~1)Gamma' (~1)Gamma' (~1)    4
##      AICc DeltaAICc      weight Deviance
## 4 855.3908  0.000000 5.535834e-01 845.2658
## 2 857.1925  1.801710 2.248779e-01 849.1093
## 5 857.2225  1.831715 2.215293e-01 845.0471
## 1 878.3273 22.936543 5.788635e-06 872.2775
## 3 879.2648 23.874000 3.622512e-06 871.1816

print("Model selection table:")

## [1] "Model selection table:"

tbDT[, modelPosition := rownames(print(tb))]

##                                model npar
## 4 alpha(~effort + site)sigma(~1)U(~site)Phi(~1)Gamma' (~1)Gamma' (~1)    5
## 2      alpha(~effort)sigma(~1)U(~site)Phi(~1)Gamma' (~1)Gamma' (~1)    4
## 5 alpha(~effort * site)sigma(~1)U(~site)Phi(~1)Gamma' (~1)Gamma' (~1)    6
## 1      alpha(~1)sigma(~1)U(~site)Phi(~1)Gamma' (~1)Gamma' (~1)    3
## 3      alpha(~site)sigma(~1)U(~site)Phi(~1)Gamma' (~1)Gamma' (~1)    4
##      AICc DeltaAICc      weight Deviance
## 4 855.3908  0.000000 5.535834e-01 845.2658
## 2 857.1925  1.801710 2.248779e-01 849.1093
## 5 857.2225  1.831715 2.215293e-01 845.0471
## 1 878.3273 22.936543 5.788635e-06 872.2775
## 3 879.2648 23.874000 3.622512e-06 871.1816

best_model <- get(paste0("mabuya_model_", which(tbDT[,modelPosition == 1])))
```

## Step 9: Interpreting Best Model Results

```
betaTable <- data.table(best_model$results$beta, keep.rownames = TRUE)
names(betaTable)[names(betaTable) == "rn"] <- "parameter"
betaTable[, Z.value := estimate / se]
betaTable[, p.value := 2 * (1 - pnorm(abs(Z.value)))]

print("Parameter estimates from best CMR model:")

## [1] "Parameter estimates from best CMR model:"

print(betaTable)

##      parameter      estimate      se      lcl      ucl      Z.value
##      <char>      <num>      <num>      <num>      <num>      <num>
## 1: alpha:(Intercept) -1.9612543 0.1597802 -2.2744234 -1.6480852 -12.274702
```



```
## 2:      alpha:effort  0.0205160 0.0040357 0.0126060 0.0284260 5.083629
## 3:      alpha:siteHPU 0.4154279 0.2066882 0.0103191 0.8205367 2.009926
## 4:      U:(Intercept) 4.2427291 0.1249664 3.9977950 4.4876632 33.950959
## 5:      U:siteHPU -2.0009952 0.2402727 -2.4719298 -1.5300606 -8.328017
##           p.value
##           <num>
## 1: 0.000000e+00
## 2: 3.702916e-07
## 3: 4.443907e-02
## 4: 0.000000e+00
## 5: 0.000000e+00
```

This table shows us that alpha (resighting rate/detection probability) is significantly **HIGHER** in HPU while U (population size) is significantly **LOWER** in HPU. So, this means that we DO see/capture more in HPU, and yet, the population size is considerably smaller. This is likely due to pressure of invasive species and the potential effects of food supplementation.

**Note:** We did not manage capture-recapture on LPU islands due to impossible logistics.

## Body Size Analysis

Using head length as the most stable morphometric variable.

### Step 1: Loading Size Data

```
mabuia_ds <- fread("data/Size_Data.csv")
```

### Step 2: Assessing Size Analysis Sampling Balance

```
pointsCounts_size <- mabuia_ds[, .N, by = "site"]
pointsCounts_size[site == "HPS", pointsArea := 5.466]
pointsCounts_size[site == "HPU", pointsArea := 11.424]
pointsCounts_size[site == "LPU", pointsArea := 1.33]
pointsCounts_size[, pointsDensity := N/pointsArea]
pointsCounts_size[, propPoints := N/(sum(pointsCounts_size[, N]))]
pointsCounts_size[, propSize := pointsArea/(sum(pointsCounts_size[, pointsArea]))]
```

```
print("Size analysis sampling distribution:")
```

```
## [1] "Size analysis sampling distribution:"
```

```
print(pointsCounts_size)
```

```
## Index: <site>
##      site      N pointsArea pointsDensity propPoints  propSize
##      <char> <int>      <num>      <num>      <num>      <num>
## 1:   HPS     58      5.466      10.611050  0.4754098 0.30000000
## 2:   HPU     44     11.424       3.851541  0.3606557 0.62700329
## 3:   LPU     20      1.330      15.037594  0.1639344 0.07299671
```

There is indeed a bit of unbalance but it is not as critical as it seems:

- **HPS:** 47% points over 30% area
- **HPU:** 36% points over 62% area
- **LPU:** 16% points over 7% area

### Step 3: Factor Level Setup

```
mabuia_ds[, site := factor(site, levels = c("LPU", "HPS", "HPU"))] # "LPU" as reference
mabuia_ds[, sex := factor(sex)] # R will pick a reference, usually "F" alphabetically
```

### Step 4: Outlier Detection

```
detect_outlier <- function(x) {
  Quantile1 <- quantile(x, probs=.25)
  Quantile3 <- quantile(x, probs=.75)
  IQR = Quantile3-Quantile1
  x > Quantile3 + (IQR*1.5) | x < Quantile1 - (IQR*1.5)
}

outliers <- detect_outlier(mabuia_ds$headLength[!is.na(mabuia_ds$headLength)])
print(paste("Number of outliers detected:", sum(outliers)))
```

```
## [1] "Number of outliers detected: 0"
```

No outliers detected, indicating good data quality.

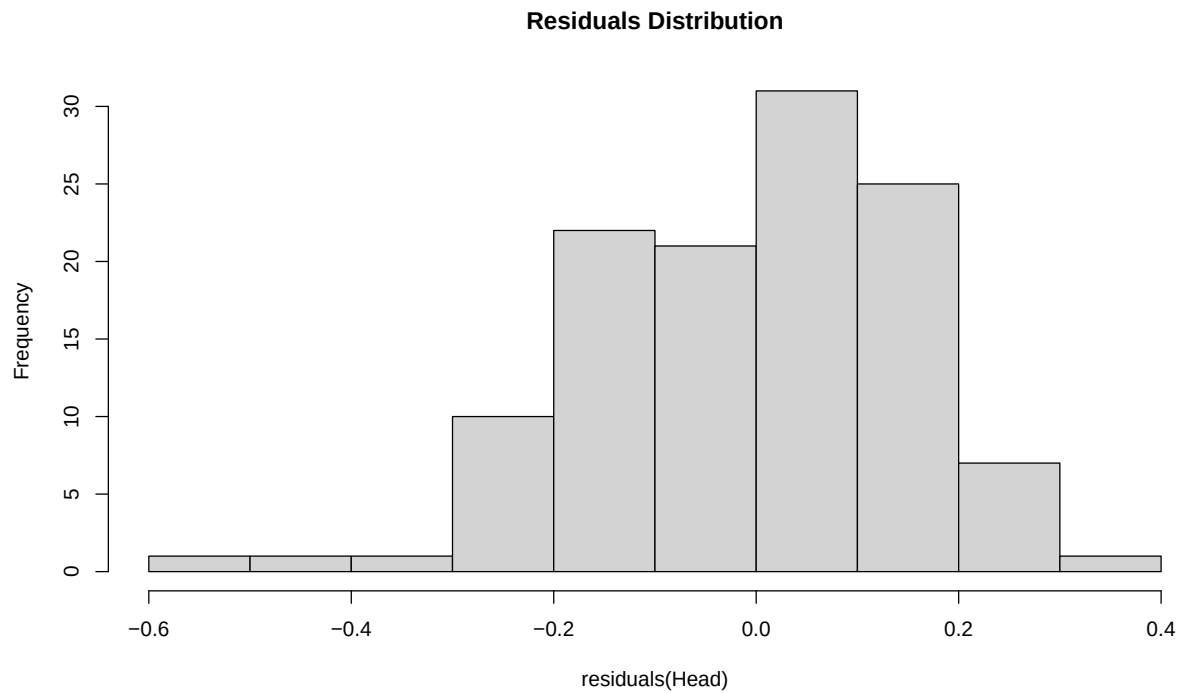
### Step 5: Head Length Model

```
Head <- glm(headLength ~ sex + site,
            family = Gamma(link = "log"),
            data = mabuia_ds)
summary(Head)
```

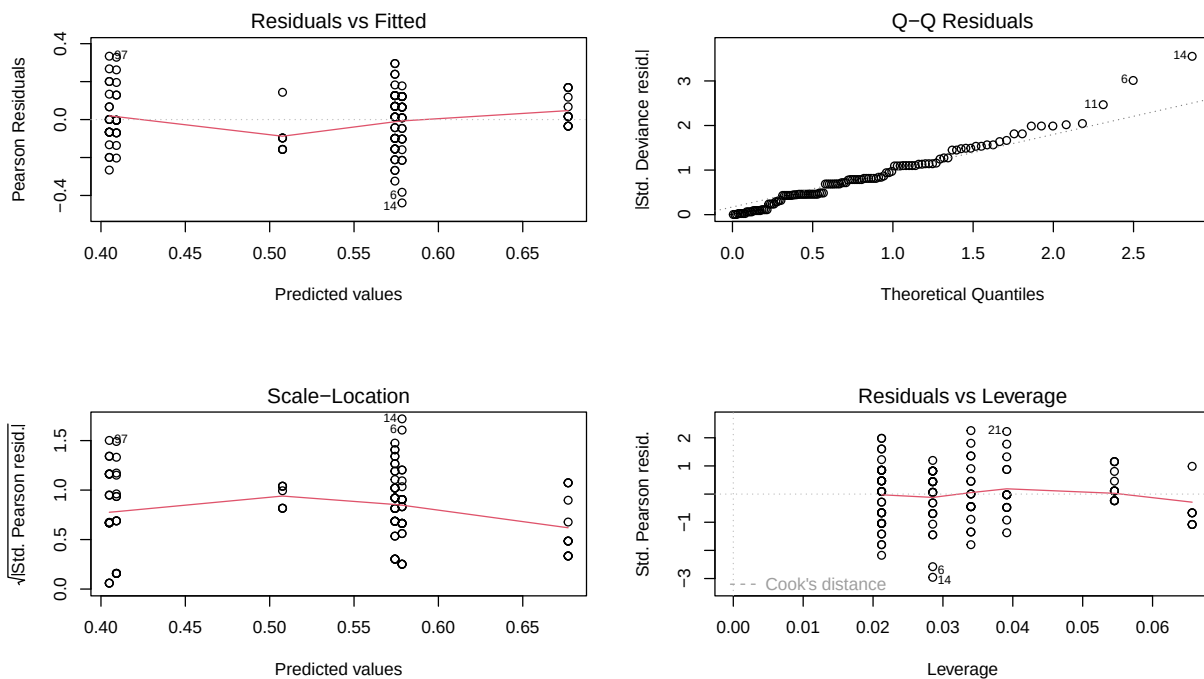
```
##
## Call:
## glm(formula = headLength ~ sex + site, family = Gamma(link = "log"),
##      data = mabuia_ds)
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.50760    0.03862  13.143 < 2e-16 ***
## sexM         0.16930    0.02902   5.833 5.01e-08 ***
## siteHPS     -0.10267    0.03909  -2.627 0.00979 **
## siteHPU     -0.09844    0.04095  -2.404 0.01780 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Gamma family taken to be 0.02271701)
##
## Null deviance: 3.7328  on 119  degrees of freedom
## Residual deviance: 2.7855  on 116  degrees of freedom
## (2 observations deleted due to missingness)
## AIC: 25.051
##
## Number of Fisher Scoring iterations: 4
```

## Step 6: Model Diagnostics

```
hist(residuals(Head), main = "Residuals Distribution")
```



```
par(mfrow=c(2,2))  
plot(Head)
```



```
par(mfrow=c(1,1))
```

## Step 7: Estimated Marginal Means for Head Length

```
emm_hF <- emmeans(Head, specs = ~ site + sex)
pairwise_head <- pairs(emm_hF, adjust = "tukey")
emm_hDT <- data.table(summary(emm_hF, type = "response"))

print("Head length estimated marginal means:")
```

```
## [1] "Head length estimated marginal means:"
```

```
print(emm_hDT)
```

|       | site   | sex    | response | SE         | df    | lower.CL | upper.CL |
|-------|--------|--------|----------|------------|-------|----------|----------|
|       | <fctr> | <fctr> | <num>    | <num>      | <num> | <num>    | <num>    |
| ## 1: | LPU    | F      | 1.661307 | 0.06416478 | 116   | 1.538960 | 1.793380 |
| ## 2: | HPS    | F      | 1.499204 | 0.04166956 | 116   | 1.418903 | 1.584050 |
| ## 3: | HPU    | F      | 1.505558 | 0.04488972 | 116   | 1.419223 | 1.597146 |
| ## 4: | LPU    | M      | 1.967770 | 0.06926544 | 116   | 1.835254 | 2.109854 |
| ## 5: | HPS    | M      | 1.775764 | 0.03898873 | 116   | 1.700197 | 1.854690 |
| ## 6: | HPU    | M      | 1.783290 | 0.04540648 | 116   | 1.695587 | 1.875529 |

```
print("Pairwise comparisons:")
```

```
## [1] "Pairwise comparisons:"
```

```
print(pairwise_head)
```

|    | contrast                                                               | estimate | SE     | df  | t.ratio | p.value |
|----|------------------------------------------------------------------------|----------|--------|-----|---------|---------|
| ## | LPU F - HPS F                                                          | 0.10267  | 0.0391 | 116 | 2.627   | 0.0991  |
| ## | LPU F - HPU F                                                          | 0.09844  | 0.0409 | 116 | 2.404   | 0.1634  |
| ## | LPU F - LPU M                                                          | -0.16930 | 0.0290 | 116 | -5.833  | <0.0001 |
| ## | LPU F - HPS M                                                          | -0.06663 | 0.0483 | 116 | -1.380  | 0.7392  |
| ## | LPU F - HPU M                                                          | -0.07086 | 0.0503 | 116 | -1.408  | 0.7219  |
| ## | HPS F - HPU F                                                          | -0.00423 | 0.0305 | 116 | -0.138  | 1.0000  |
| ## | HPS F - LPU M                                                          | -0.27197 | 0.0491 | 116 | -5.542  | <0.0001 |
| ## | HPS F - HPS M                                                          | -0.16930 | 0.0290 | 116 | -5.833  | <0.0001 |
| ## | HPS F - HPU M                                                          | -0.17353 | 0.0427 | 116 | -4.061  | 0.0012  |
| ## | HPU F - LPU M                                                          | -0.26774 | 0.0501 | 116 | -5.347  | <0.0001 |
| ## | HPU F - HPS M                                                          | -0.16507 | 0.0415 | 116 | -3.973  | 0.0017  |
| ## | HPU F - HPU M                                                          | -0.16930 | 0.0290 | 116 | -5.833  | <0.0001 |
| ## | LPU M - HPS M                                                          | 0.10267  | 0.0391 | 116 | 2.627   | 0.0991  |
| ## | LPU M - HPU M                                                          | 0.09844  | 0.0409 | 116 | 2.404   | 0.1634  |
| ## | HPS M - HPU M                                                          | -0.00423 | 0.0305 | 116 | -0.138  | 1.0000  |
| ## |                                                                        |          |        |     |         |         |
| ## | Results are given on the log (not the response) scale.                 |          |        |     |         |         |
| ## | P value adjustment: tukey method for comparing a family of 6 estimates |          |        |     |         |         |

## Step 8: Testing for Interaction Effects

```
Head_Interaction <- glm(headLength ~ sex * site,
                        family = Gamma(link = "log"),
                        data = mabuia_ds)
summary(Head_Interaction)
```

```
##
## Call:
## glm(formula = headLength ~ sex * site, family = Gamma(link = "log"),
##      data = mabuia_ds)
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.414944   0.056381   7.360 3.07e-11 ***
## sexM          0.308509   0.069931   4.412 2.34e-05 ***
## siteHPS      -0.002486   0.065954  -0.038  0.9700
## siteHPU       0.025459   0.068280   0.373  0.7099
## sexM:siteHPS -0.150422   0.081438  -1.847  0.0673 .
## sexM:siteHPU -0.188238   0.084841  -2.219  0.0285 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Gamma family taken to be 0.02225138)
##
## Null deviance: 3.7328  on 119  degrees of freedom
## Residual deviance: 2.6737  on 114  degrees of freedom
## (2 observations deleted due to missingness)
## AIC: 24.116
##
## Number of Fisher Scoring iterations: 4
emm_interaction <- emmeans(Head_Interaction, specs = ~ sex * site)
pairs_interaction_head <- pairs(emm_interaction, adjust = "tukey")
print("Interaction model pairwise comparisons:")

## [1] "Interaction model pairwise comparisons:"
print(pairs_interaction_head)

## contrast      estimate      SE df t.ratio p.value
## F LPU - M LPU -0.30851 0.0699 114 -4.412 0.0003
## F LPU - F HPS  0.00249 0.0660 114  0.038 1.0000
## F LPU - M HPS -0.15560 0.0612 114 -2.541 0.1210
## F LPU - F HPU -0.02546 0.0683 114 -0.373 0.9990
## F LPU - M HPU -0.14573 0.0633 114 -2.303 0.2013
## M LPU - F HPS  0.31099 0.0537 114  5.792 <0.0001
## M LPU - M HPS  0.15291 0.0478 114  3.201 0.0214
## M LPU - F HPU  0.28305 0.0565 114  5.008 <0.0001
## M LPU - M HPU  0.16278 0.0504 114  3.233 0.0195
## F HPS - M HPS -0.15809 0.0417 114 -3.788 0.0033
## F HPS - F HPU -0.02795 0.0515 114 -0.542 0.9943
## F HPS - M HPU -0.14822 0.0447 114 -3.318 0.0151
## M HPS - F HPU  0.13014 0.0453 114  2.872 0.0537
## M HPS - M HPU  0.00987 0.0373 114  0.264 0.9998
## F HPU - M HPU -0.12027 0.0480 114 -2.504 0.1317
##
## Results are given on the log (not the response) scale.
## P value adjustment: tukey method for comparing a family of 6 estimates
```

## Step 9: Ecological Interpretation of Size Patterns

```
dt <- as.data.table(emm_interaction)

# Compare sex ratios across sites
male_props <- table(mabuia_ds[sex == "M", site])/NROW(mabuia_ds[sex == "M",])
female_props <- table(mabuia_ds[sex == "F", site])/NROW(mabuia_ds[sex == "F",])

print("Proportion of males by site:")

## [1] "Proportion of males by site:"
print(male_props)

##
##      LPU      HPS      HPU
## 0.1645570 0.4936709 0.3417722
print("Proportion of females by site:")

## [1] "Proportion of females by site:"
print(female_props)

##
##      LPU      HPS      HPU
## 0.1627907 0.4418605 0.3953488
```

## Injury Analysis (Autotomy)

### Step 1: Creating Autotomy Variable

```
mabuia_ds[, has_autotomy := ifelse(autotomy == "", "No", "Yes")]
```

### Step 2: Autotomy Proportions and Statistical Test

```
autotomy_table_3way <- table(mabuia_ds$site, mabuia_ds$has_autotomy)
propTable <- prop.table(autotomy_table_3way, margin = 1)
chi_test_autotomy_3way <- chisq.test(autotomy_table_3way)
correctedProportion <- round(propTable*100, 2)
colnames(correctedProportion) <- c("No autotomy signs", "Autotomy signs")

print("Autotomy proportions by site (%):")

## [1] "Autotomy proportions by site (%):"
print(correctedProportion)

##
##      No autotomy signs Autotomy signs
##  LPU      85.00      15.00
##  HPS      70.69      29.31
##  HPU      75.00      25.00
print("Chi-square test results:")

## [1] "Chi-square test results:"
```

```
print(chi_test_autotomy_3way)
```

```
##  
## Pearson's Chi-squared test  
##  
## data: autotomy_table_3way  
## X-squared = 1.613, df = 2, p-value = 0.4464
```

Autotomy was proportionally more frequent in the HPS than in HPU and roughly twice that observed on the secondary islands, suggesting elevated predation pressure on the main island. However, these differences were not statistically significant. Our power analyses indicate that this outcome is largely due to limited sample sizes—particularly on the secondary islands—rather than the absence of a biological effect, highlighting the need for expanded sampling to better resolve these patterns.

## Combined Size and Injury Visualization

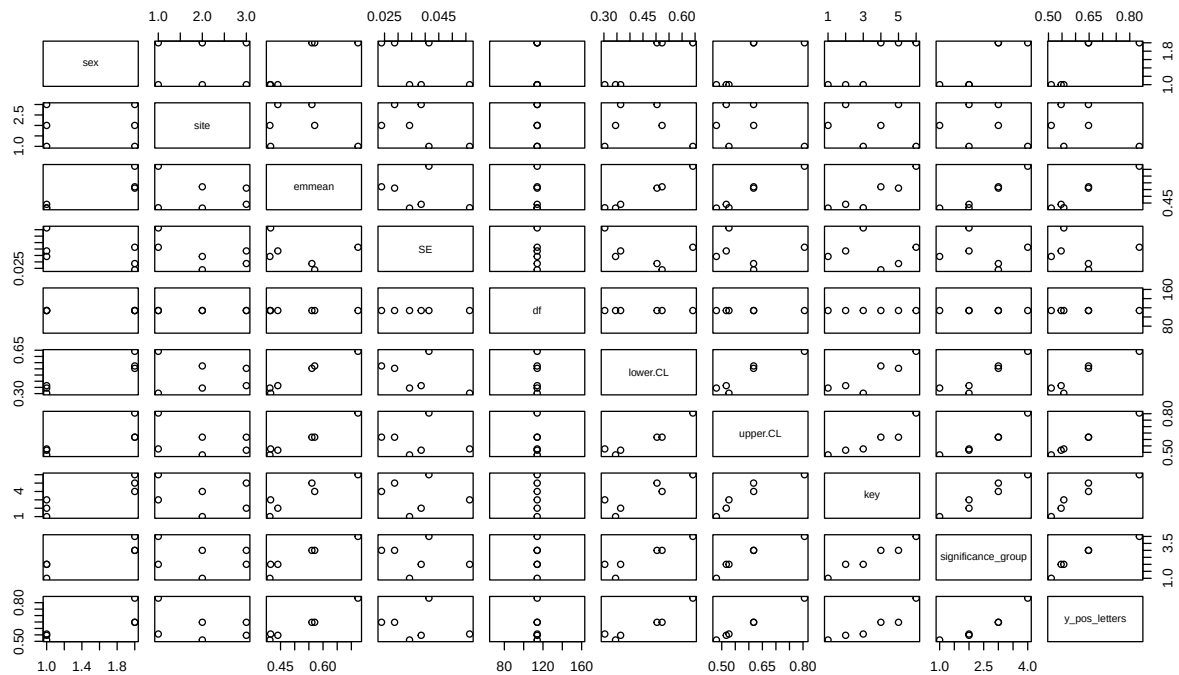
### Step 1: Preparing Interaction Data for Plotting

```
dt_df <- as.data.frame(dt)  
dt_df$site <- factor(dt_df$site, levels = c("LPU", "HPS", "HPU"))  
  
# Get significance testing for interaction model  
pairs_interaction <- contrast(emm_interaction, method = "pairwise", adjust = "tukey")  
summary_pw_int <- summary(pairs_interaction)  
p_values_int <- summary_pw_int$p.value  
print(p_values_int)
```

```
## [1] 3.312743e-04 1.000000e+00 1.210025e-01 9.990378e-01 2.012888e-01  
## [6] 9.272989e-07 2.142024e-02 2.956024e-05 1.949409e-02 3.262026e-03  
## [11] 9.942631e-01 1.505577e-02 5.373153e-02 9.998207e-01 1.316787e-01
```

### Step 2: Processing Significance Letters for Interaction

```
contrast_names_raw_int <- as.character(summary_pw_int$contrast)  
p_value_names <- sapply(contrast_names_raw_int, function(contrast) {  
  parts <- strsplit(contrast, " - ")[[1]]  
  key1 <- gsub(" ", "", parts[1])  
  key2 <- gsub(" ", "", parts[2])  
  return(paste(key1, key2, sep = "-"))  
})  
names(p_values_int) <- p_value_names  
  
letters_result_int <- multcompLetters(p_values_int)  
significance_letters_int <- letters_result_int$Letters  
  
dt_df$key <- paste0(dt_df$sex, dt_df$site)  
dt_df$significance_group <- significance_letters_int[dt_df$key]  
dt_df$y_pos_letters <- dt_df$upper.CL + 0.03  
plot(dt_df)
```



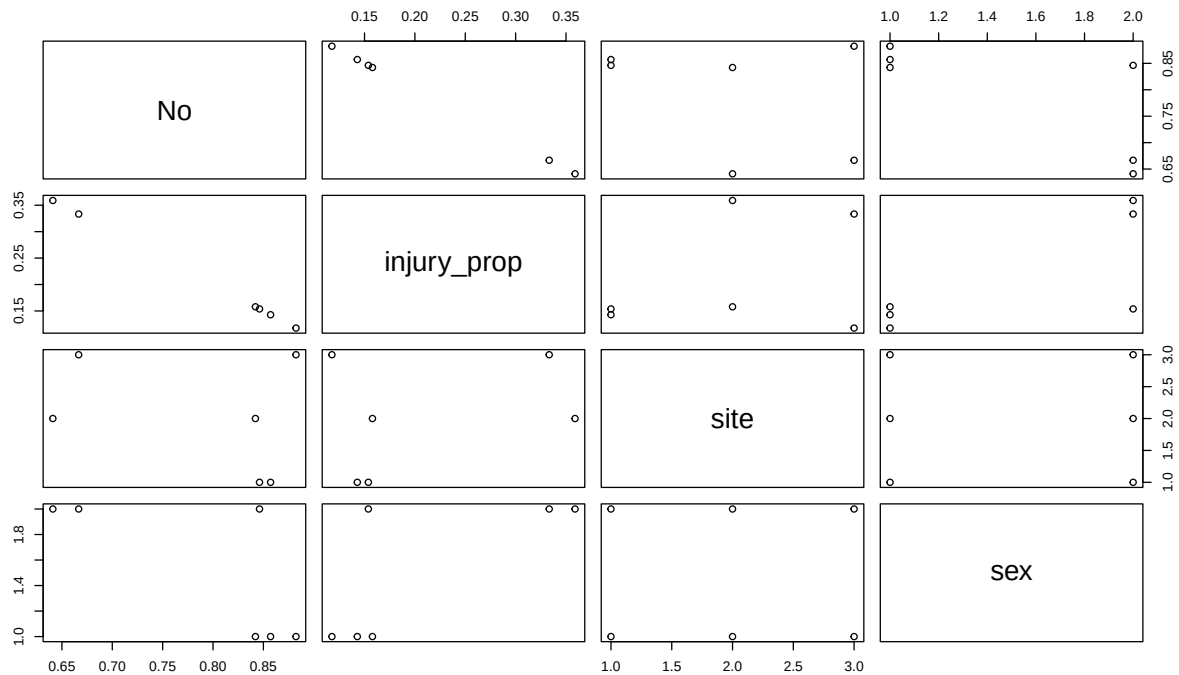
### Step 3: Preparing Autotomy Data by Sex

```
# For males
males_data <- mabuia_ds[mabuia_ds$sex == 'M', ]
prop_table_males <- prop.table(table(males_data$site, males_data$has_autotomy), margin = 1)
males_autotomy_df <- as.data.frame.matrix(prop_table_males)
names(males_autotomy_df)[names(males_autotomy_df) == "Yes"] <- "injury_prop"
males_autotomy_df$site <- rownames(males_autotomy_df)
males_autotomy_df$sex <- "Males"

# For females
females_data <- mabuia_ds[mabuia_ds$sex == 'F', ]
prop_table_females <- prop.table(table(females_data$site, females_data$has_autotomy), margin = 1)
females_autotomy_df <- as.data.frame.matrix(prop_table_females)
names(females_autotomy_df)[names(females_autotomy_df) == "Yes"] <- "injury_prop"
females_autotomy_df$site <- rownames(females_autotomy_df)
females_autotomy_df$sex <- "Females"

# Combine
autotomy_df_final <- rbind(males_autotomy_df, females_autotomy_df)
autotomy_df_final$site <- factor(autotomy_df_final$site, levels = c("LPU", "HPS", "HPU"))
autotomy_df_final$sex <- factor(autotomy_df_final$sex, levels = c("Females", "Males"))
plot(autotomy_df_final)
```





#### Step 4: Formatting for Unified Plot

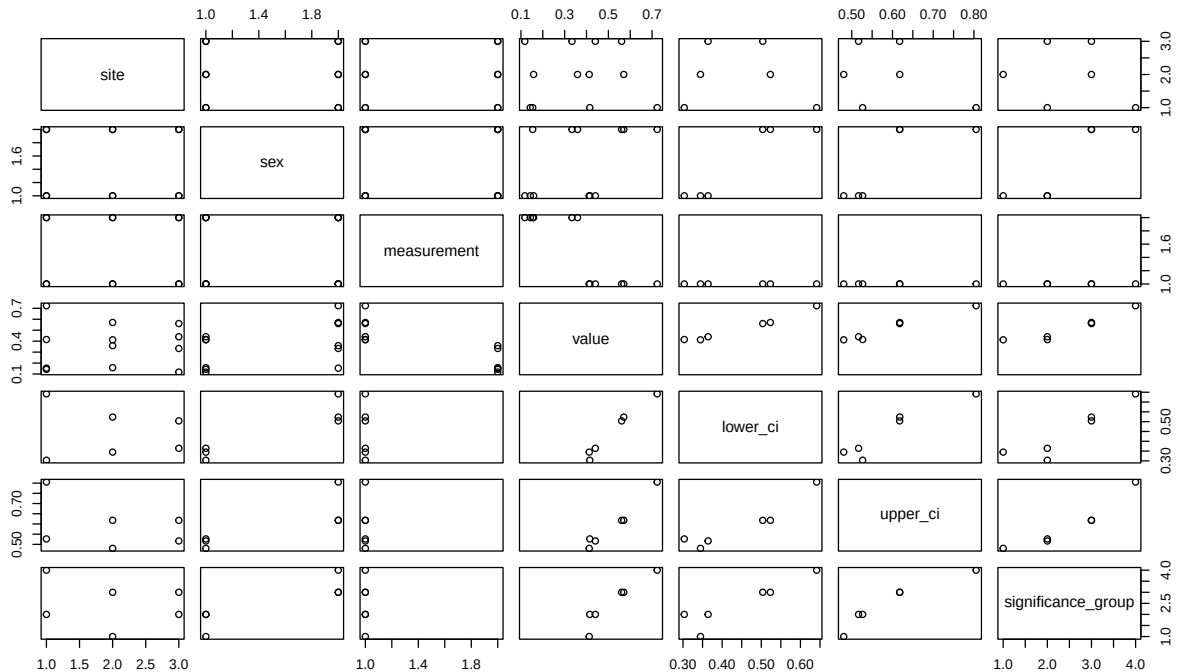
```
dt_df$sex <- ifelse(dt_df$sex == "F", "Females", "Males")
dt_df$sex <- factor(dt_df$sex, levels = c("Females", "Males"))

# Create unified plotting dataset
# Select and rename columns from head length data
plot_data_head <- data.frame(
  site = dt_df$site,
  sex = dt_df$sex,
  measurement = "Head Length (cm)",
  value = dt_df$emmean,
  lower_ci = dt_df$lower.CL,
  upper_ci = dt_df$upper.CL,
  significance_group = dt_df$significance_group
)

# Select and rename columns from injury data
plot_data_injury <- data.frame(
  site = autotomy_df_final$site,
  sex = autotomy_df_final$sex,
  measurement = "Proportion Injury",
  value = autotomy_df_final$injury_prop,
  lower_ci = NA, # No error bars for injury rate
  upper_ci = NA,
  significance_group = NA # No significance letters for injury rate
)

plot_data_unified <- rbind(plot_data_head, plot_data_injury)
```

```
plot(plot_data_unified)
```



## Creating the Final Size and Injury Plot

```
body_size_plot <- ggplot(
  plot_data_unified,
  aes(x = site, y = value, fill = site, color = site)
) +
  # --- Geoms that use SUBSETS of the data ---
  # Add bars ONLY for the injury rate data
  geom_bar(
    data = plot_data_unified[plot_data_unified$measurement == "Proportion Injury", ],
    stat = "identity", color = "black", alpha = 0.8
  ) +
  # Add error bars ONLY for the head length data
  geom_errorbar(
    data = plot_data_unified[plot_data_unified$measurement == "Head Length (cm)", ],
    aes(ymin = lower_ci, ymax = upper_ci), width = 0.25, linewidth = 1
  ) +
  # Add points ONLY for the head length data
  geom_point(
    data = plot_data_unified[plot_data_unified$measurement == "Head Length (cm)", ],
    size = 4, stroke = 1.5
  ) +
  # Add text ONLY for the head length data
  geom_text(
    data = plot_data_unified[plot_data_unified$measurement == "Head Length (cm)", ],
    aes(y = upper_ci + 0.1, label = significance_group), # Increase offset for % scale
    color = "black", size = 6, fontface = "bold"
```

```

) +

# --- Faceting ---
# Facet by both sex and measurement, and allow y-axes to be independent
facet_grid(
  measurement ~ sex,
  scales = "free_y"
) +

# --- Scales and Labels ---
scale_color_viridis_d(option = "D") +
scale_fill_viridis_d(option = "D") +

# --- Theming ---
theme_bw(base_size = 14) +
theme(
  legend.position = "none",
  plot.title = element_text(face = "bold", size = 16),
  strip.text = element_text(size = 12, face = "bold"),
  strip.placement = "outside", # Ensures strips are next to the axis
  axis.title = element_text(face = "bold"),
  panel.grid.major.x = element_blank(),
  panel.grid.minor = element_blank(),
  # Add a bit more space between facets
  panel.spacing = unit(1.0, "lines"),
  axis.title.x = element_blank(),
  axis.title.y = element_blank(),
)

```

## Summarizing the findings:

### Results from point counts:

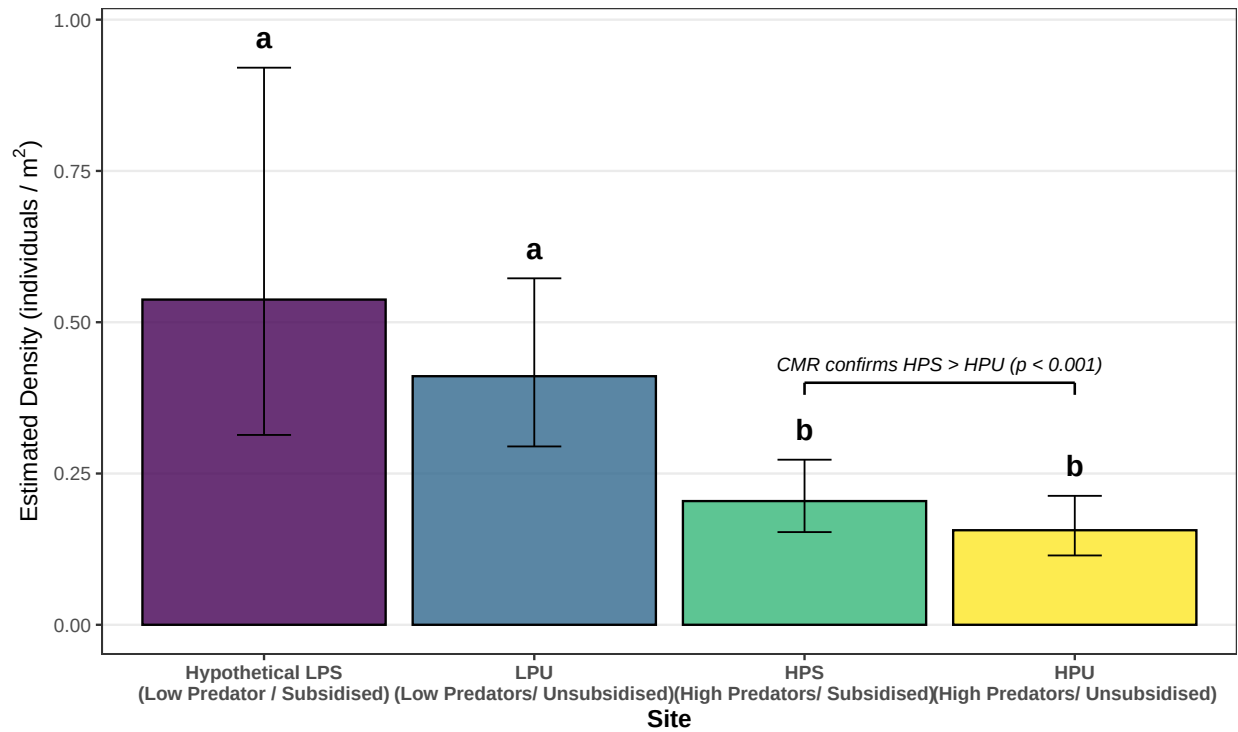
```
summary(emm_density_hierarchical, infer = TRUE)
```

```

## InvasiveSpeciesPresence FoodSupplementation response      SE  df asymp.LCL
## 0                        0          0.411 0.0696 Inf      0.295
## 1                        0          0.156 0.0247 Inf      0.115
## 0                        1          0.537 0.1480 Inf      0.314
## 1                        1          0.204 0.0301 Inf      0.153
## asymp.UCL null z.ratio p.value
##    0.573    1  -5.252 <0.0001
##    0.213    1 -11.730 <0.0001
##    0.921    1  -2.261  0.0238
##    0.273    1 -10.782 <0.0001
##
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
## Tests are performed on the log scale

```

```
density_plot
```



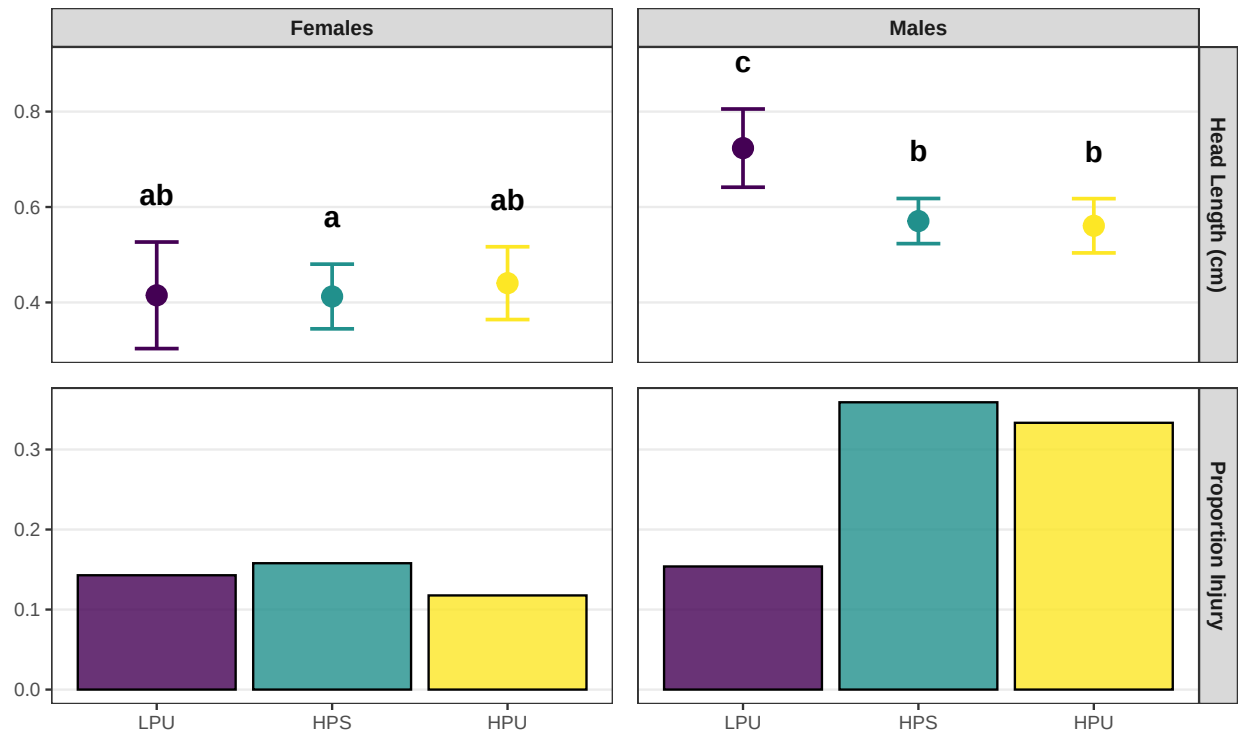
## Results from CMR:

```
print(betaTable)
```

```
##           parameter  estimate      se      lcl      ucl  Z.value
##           <char>      <num>      <num>      <num>      <num>      <num>
## 1: alpha:(Intercept) -1.9612543 0.1597802 -2.2744234 -1.6480852 -12.274702
## 2:   alpha:effort    0.0205160 0.0040357  0.0126060  0.0284260   5.083629
## 3:   alpha:siteHPU   0.4154279 0.2066882  0.0103191  0.8205367   2.009926
## 4:   U:(Intercept)  4.2427291 0.1249664  3.9977950  4.4876632  33.950959
## 5:   U:siteHPU     -2.0009952 0.2402727 -2.4719298 -1.5300606  -8.328017
##           p.value
##           <num>
## 1: 0.000000e+00
## 2: 3.702916e-07
## 3: 4.443907e-02
## 4: 0.000000e+00
## 5: 0.000000e+00
```

## Results from size analysis:

```
body_size_plot
```



## Results from sublethal injuries:

```
print(correctedProportion)
```

```
##
##      No autotomy signs Autotomy signs
##  LPU          85.00      15.00
##  HPS          70.69      29.31
##  HPU          75.00      25.00
```

```
print(chi_test_autotomy_3way)
```

```
##
##  Pearson's Chi-squared test
##
## data:  autotomy_table_3way
## X-squared = 1.613, df = 2, p-value = 0.4464
```

## Saving the plots for manuscript

```
ggsave(
  filename = "outputs/Density_Plot.png", # The name of the output file
  plot = density_plot,                  # The plot object to save
  width = 12,                           # Width of the image in inches
  height = 6,                           # Height of the image in inches
  dpi = 300,                            # Dots per inch (standard for publication)
  bg = "white"                          # Set a white background
)
```

```

# Save as a high-resolution TIFF file
ggsave(
  filename = "outputs/Density_Plot.tiff",
  plot = density_plot,
  width = 12,
  height = 6,
  dpi = 300
)

ggsave(
  filename = "outputs/Size_Injury.png",
  plot = body_size_plot,
  width = 8,    # Keep it reasonably wide for the two columns
  height = 7,   # A bit taller to accommodate the stacked facets
  dpi = 300,
  bg = "white"
)

ggsave(
  filename = "outputs/Size_Injury.tiff",
  plot = body_size_plot,
  width = 8,    # Keep it reasonably wide for the two columns
  height = 7,   # A bit taller to accommodate the stacked facets
  dpi = 300,
  bg = "white"
)

```